T3000 API & Protocol

Message: GET_PANEL_DATA · built Nov 13, 2026

Message: GET_PANEL_DATA

GET_PANEL_DATA loads all cached data for a specific panel. This is the fastest way to get a complete snapshot of a device's current state without querying the device directly.

Common Uses:

- Initial panel view load
- Refreshing entire panel data
- Offline data access

Overview

Message Type: GET_PANEL_DATA **File Location:** `BacnetWebView.cpp` line 893 **Purpose:** Load all cached panel data from local storage

Request Format

JSON

```
{
  "action": "GET_PANEL_DATA",
  "source": 1,
  "panelId": 0
}
```

Parameters

Parameter	Type	Required	Description
`action`	String	✅	Must be "GET_PANEL_DATA"
`source`	Integer	✅	Message source: 0=T3000, 1=WebUI
`panelId`	Integer	✅	Panel number (0-254)

Response Format

JSON

```
{
  "action": "GET_PANEL_DATA_RES",
  "data": {
    "panel_info": {
      "panel_number": 0,
      "serial_number": 12345,
      "object_instance": 123456,
      "panel_name": "Controller 1",
      "panel_type": 19,
      "online_time": 1704672000
    },
    "inputs": [...],
    "outputs": [...],
    "variables": [...],
    "programs": [...],
    "controllers": [...],
    "monitors": [...],
    "schedules": [...],
    "holidays": [...]
  }
}
```

Implementation

Code Location

C++

```
case WEBVIEW_MESSAGE_TYPE::GET_PANEL_DATA:
{
    tempjson["action"] = "GET_PANEL_DATA_RES";
    int npanel_id = json.get("panelId", Json::nullValue).asInt();

    if (npanel_id == 0)
        npanel_id = bac_gloab_panel;

    int nret = LoadOnlinePanelData(npanel_id);

    if (nret < 0)
    {
        CString temp_message;
        temp_message.Format(_T("No cached data found for panel %d"), npanel_id);
        WrapErrorMessage(builder, tempjson, outmsg, temp_message);
        break;
    }

    // Populate all data arrays...
    break;
}
```

Data Loading

1. **Check cache** - Reads from `DatabaseBak/` folder
2. **Load file** - Deserializes saved panel data
3. **Populate arrays** - Fills global arrays (g_Input_data, g_Output_data, etc.)
4. **Build JSON** - Creates response with all data

Performance

Scenario	Time
Cache hit	< 100ms
No cache	Returns error immediately
Large panel (all data)	100-500ms

Error Handling

Cache Miss:

```JSON
{
  "action": "ERROR_RES",
  "error": {
    "code": -3,
    "message": "No cached data found for panel 0"
  }
}
```

Use Cases

Initial Load

Load all data when opening panel view for the first time.

Offline Access

Access cached data when device is offline or unreachable.

Quick Refresh

Get current cached state without network delay.

See Also

- GET_WEBVIEW_LIST - Read fresh data from device
- Control Messages Index